

The Generational Economy & National Transfer Accounts

Population Economics – Lecture 5

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This block: Lectures 5–8

The fiscal side of population aging

L5 The generational economy & National Transfer Accounts

L6 Pensions & labor supply

L7 Aging & healthcare I: the red herring hypothesis

L8 Aging & healthcare II: steepening & forecasting

Learning outcomes

By the end of this lecture you should be able to:

- Explain how economic resources (earnings, transfers, savings) flow across age groups.
- Interpret the National Transfer Accounts flow identity.
- Compute and interpret the support ratio.
- Assess the empirical evidence on whether population aging slows economic growth.

Readings

Part I — the lifecycle

Lee, R., & Mason, A. (eds.) (2011). *Population Aging and the Generational Economy: A Global Perspective*, Ch. 3: Introducing age into national accounts (pp. 55–78). Edward Elgar.

The economic lifecycle, and the accounting that goes with it.

Part II — projecting it forward

Acemoglu, D., & Restrepo, P. (2017). Secular stagnation? The effect of aging on economic growth in the age of automation. *American Economic Review*, 107(5), 174–179.

Does aging slow growth? Across countries — **no**.

Maestas, N., Mullen, K. J., & Powell, D. (2023). The effect of population aging on economic growth, the labor force, and productivity. *American Economic Journal: Macroeconomics*, 15(2), 306–332.

The same question inside one country — **yes**.

Key questions ($\approx$ 3 minutes)

- When do individuals earn income across the lifecycle?
- When do individuals consume across the lifecycle?
- How are the two gaps bridged?
- What does the population age structure imply for macroeconomic outcomes?

PART I

Who pays for the lifecycle deficit?

Consumption does not track labor income. At almost every age there is a gap, and somebody closes it — the family, the state, or your own past savings.

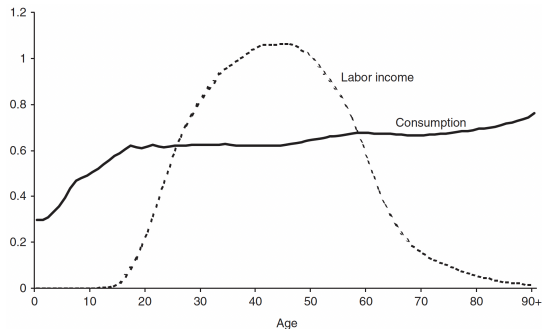
Lee & Mason (2011)

What is the “generational economy”?

Definition: Generational economy (Lee and Mason, 2011)

1. “The social institutions and **economic mechanisms** used by each generation or age group to **produce, consume, share, and save** resources”
2. “The economic **flows across generations or age groups** that characterize the generational economy”
3. “**Explicit and implicit contracts** that govern intergenerational flows”

The economic lifecycle



1. Consumption does not track income.
2. Two deficits, one surplus. The deficits **must be financed somehow**.
3. This is obtained across 23 economies, rich and poor, with very different welfare states.

Question (≈ 3 minutes)

What should we expect from population aging?

Major consumption items across the lifecycle

Item	When in life	Possibilities
Childcare	ages 0–6	Parents? Grandparents? The state?
Education	ages 6–25	Parents? The state? Student debt?
Old age	ages 65+	Public pension? Family support? Own assets?
Healthcare	all ages	Public? Private insurance? Employer?

Same problem, but different solutions.

Think about Denmark and the United States.

The flow identity

For a person of age x , in a given year, resources in must equal resources out:

$$\underbrace{Y^\ell(x)}_{\text{labor income}} + \underbrace{Y^A(x)}_{\text{asset income}} + \underbrace{\tau^+(x)}_{\text{transfers in}} = \underbrace{C(x)}_{\text{consumption}} + \underbrace{S(x)}_{\text{saving}} + \underbrace{\tau^-(x)}_{\text{transfers out}}$$

Rearranging:

$$\underbrace{C(x) - Y^\ell(x)}_{\text{Lifecycle deficit}} = \underbrace{\tau^+(x) - \tau^-(x)}_{\text{Net transfers}} + \underbrace{Y^A(x) - S(x)}_{\text{Asset-based reallocations}}$$

Age reallocations

- This is an **accounting identity**, nothing is assumed about behaviour.
- A **positive** number means a deficit that must be financed; a negative number is a surplus.
- At the individual level it is simply a **budget constraint**; it holds in aggregate.

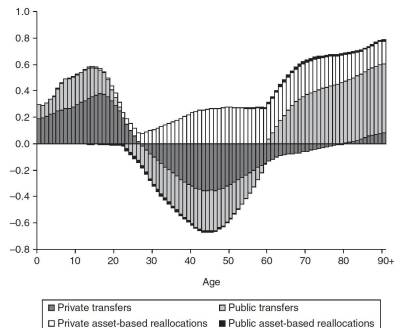
Clarifying the terms

	Transfers $\tau(x)$ one-way: nothing given in exchange	Asset-based $Y^A(x) - S(x)$ moving your own resources across time
Public	In: public healthcare, education Out: taxes	In: Dividends, rent Out: public saving
Private	In: support received from family Out: support given to family	In: asset income, rent Out: saving

Question (≈ 2 minutes)

How would you classify SU? And a student loan?

How the gap is financed: age reallocations

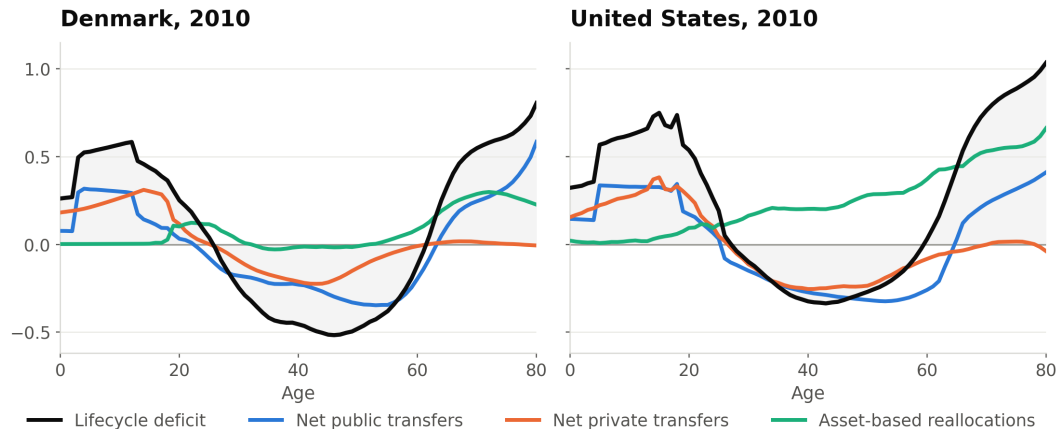


- **Children:** inflow, **dominated by private transfers**; public transfers (education) rise then fall.
- **Prime age (27–58):** net payers on *both* transfer channels, **private and public**.
- **Old age (60+):** inflow again, but **public transfers and private assets**.

Takeaway

The asymmetry between children and the elderly is exactly why population aging is a **fiscal** problem.

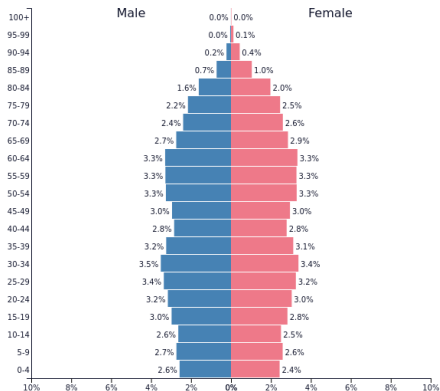
Denmark and the United States



Question (≈ 3 minutes)

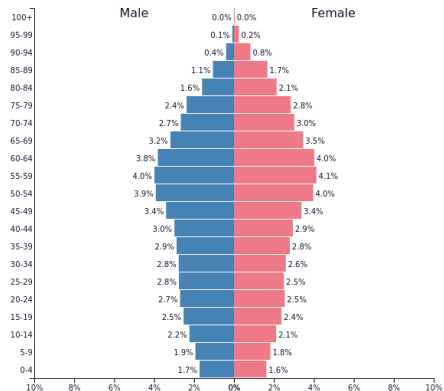
What differs between Denmark and the US, and what does not?

Visualising age structure



PopulationPyramid.net

Denmark - 2026
Population: **6,023,520**



PopulationPyramid.net

Italy - 2026
Population: **58,926,166**

Question ($\approx$ 2 minutes)

What does the figure imply?

From profiles to aggregates

To get a national total, weight each age by how many people are in it:

$$\text{Aggregate}_t = \sum_a \underbrace{x(a, t_0)}_{\text{per capita profile}} \underbrace{N(a, t)}_{\text{n. people aged } a}.$$

Do this for **labor income** and for **consumption**, and take the ratio of the two:

$$SR_t = \frac{\sum_a y_l(a, t_0) N(a, t)}{\sum_a c(a, t_0) N(a, t)} = \frac{\text{effective producers}}{\text{effective consumers}}.$$

- **Not a head count:** we weight each person by what they actually earn and consume.
- **Higher is more favourable:** labor income covers more of consumption.
- **Profiles are frozen at t_0 :** only $N(a, t)$ moves, so changes in SR are purely demographic.

What a falling support ratio looks like

Age	$y_I(a)$	$c(a)$	N_{2025}	N_{2075}	
0–19	0.0	0.5	20	20	
20–64	1.0	1.0	60	40	
65+	0.1	1.5	20	40	
Total			100	100	
Effective producers			62	44	
Effective consumers			100	110	
Support ratio			0.62	0.40	–35%

The same 100 people throughout — 20 of them simply move from work into retirement.

Question

What are we assuming here?

PART II

What breaks when we project it forward?

$$\text{Aggregate}_t = \sum_a \mathbf{x}(\mathbf{a}, \mathbf{t}_0) N(a, t)$$

The accounts are frozen at t_0 . Part II asks what that assumption costs, and whether the slowdown it predicts is in the data at all.

Guvenen et al. (2022) · Acemoglu & Restrepo (2017) · Maestas et al. (2023)

Where we left off

- Consumption does not track labor income; **two lifecycle deficits** closed by transfers or by assets.
- Aggregates are profiles $\times$ population: $\text{Aggregate}_t = \sum_a x(a, t_0)N(a, t)$.
- **The support ratio** applies this to y_l and c .

So what are we assuming?

Freezing t_0 says the **profiles themselves never change**. Two reasons they might:

- **Cohorts differ**. The profile is a *cross-section*: today's 25- and 65-year-olds are different people.
- **Profiles respond**. A changing age structure can itself change behaviour and technology.

Guvenen et al. (2022): are cohorts alike?

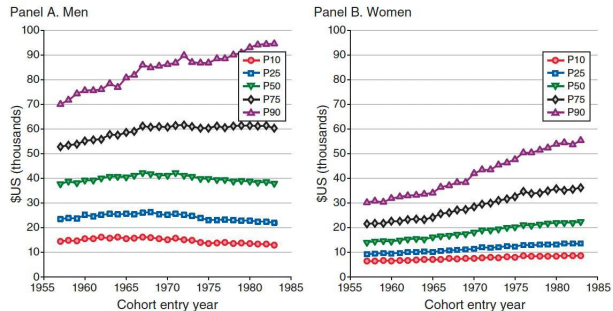


FIGURE 3. SELECT PERCENTILES OF LIFETIME EARNINGS BY COHORT AND GENDER

Notes: An observation represents a select quantile of the lifetime earnings distribution of a cohort that entered the labor market in a given year for the baseline sample (see Section IC). Panel A displays the distribution for men and panel B, for women. Values are displayed in thousands of 2013 US dollars and deflated using the PCE.

Question (≈ 3 minutes)

How do these results impact the NTA?

Acemoglu and Restrepo (2017): does technology respond to aging?

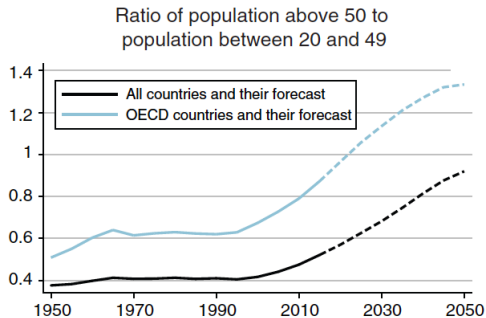


FIGURE 1. AGING FROM 1950 TO 2015 AND PROJECTIONS UNTIL 2050 (*From UN data*)

Note: Aging is measured by the ratio of the population above 50 years old to the population between 20 and 49.

Every paper on aging starts by showing that **aging** is real.

Acemoglu and Restrepo (2017) (cont'd): Aging and growth across countries

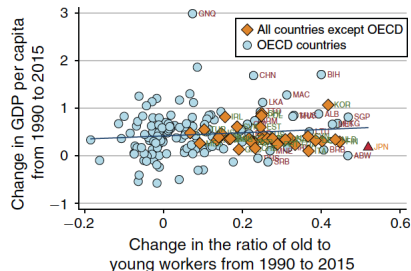


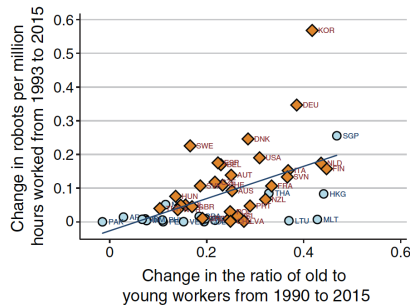
FIGURE 2. CORRELATION BETWEEN AGING AND GROWTH IN GDP PER CAPITA (*In Constant Dollars*)

Note: Aging is defined as the change in the ratio of the population above 50 years old to the population between 20 and 49.

Source: GDP per capita data from the Penn World Tables

The puzzle: an older population earns less labor income per head, so slower growth $\Rightarrow$ we expect a **negative** slope.

Acemoglu and Restrepo (2017) (cont'd): robots



Maestas, Mullen and Powell (2023): the other answer

Same question, but **one country instead of many**: US states, 1980–2010.

- **What we gain.** States share a currency, a legal system, institutions and a technology frontier. The thousand ways in which countries differ are gone.
- **What we lose.** If aging induces automation and the response is common to every state, it would be absorbed by this design.

Maestas et al. (2023) (cont'd): aging in the United States

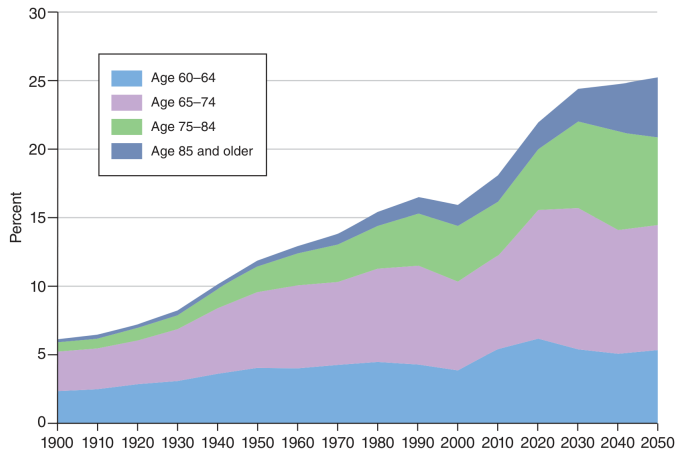
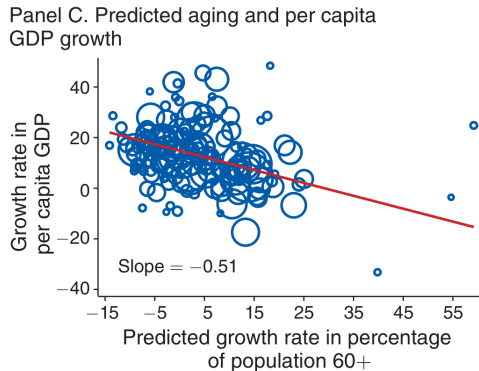


FIGURE 1. PERCENT OF UNITED STATES POPULATION AGE 60+: ACTUAL AND PROJECTED—1900–2050

The same starting point: **aging is real.**

Maestas et al. (2023) (cont'd): aging and growth across states



Within one country the sign is the one we expected.

Maestas et al. (2023) (cont'd): where the effect comes from

TABLE 2—DECOMPOSING MAIN EFFECT

Instrument length: Dependent variable:	10-year lag			
	$\Delta \ln(\text{GDP}/N)$ (1)	$\Delta \ln(\text{GDP}/\text{hours})$ (2)	$\Delta \ln(H/L)$ (3)	$\Delta \ln(L/N)$ (4)
$\Delta \ln\left(\frac{A}{N}\right)$	-0.545 (0.173)	-0.343 (0.151)	-0.031 (0.033)	-0.172 (0.047)

A 10% rise in the share aged 60+ costs **5.5%** of GDP per capita over a decade.

Columns (2)–(4) sum to column (1): **two-thirds is labor productivity**, only one-third is having fewer workers.

Question ($\approx$ 3 minutes)

How does this growth slowdown affect the NTA?

So who is right?

	Acemoglu & Restrepo (2017)	Maestas et al. (2023)
Finding	Aging $\Rightarrow$ no negative effect on growth	Aging $\Rightarrow$ -5.5% growth per decade
Variation	Across countries, 1990–2015	Across US states, 1980–2010
Mechanism	Technology offsets scarce labor	Lower productivity and labor supply
Weakness	Countries differ in a thousand ways	National tech response is invisible

Summary

Part I — who pays for the lifecycle deficit

1. We consume at all ages but produce only in the middle of life. Two **lifecycle deficits** — youth and old age — must be financed.
2. The **NTA flow identity** says there are exactly two ways to finance them:

$$C(x) - Y^{\ell}(x) = \underbrace{\tau(x)}_{\text{transfers}} + \underbrace{Y^A(x) - S(x)}_{\text{assets}}$$

each splitting into public and private. Which channel dominates is a matter of institutions, and differs sharply across countries.

3. The **support ratio** collapses profiles $\times$ population into one number, and so isolates **pure demography**.

Part II — what breaks when we project it forward

4. It isolates demography *at the price of* freezing the profiles at t_0 — and there are two reasons they move: **cohorts differ** and **profiles respond**.
5. Whether aging **slows growth** is genuinely contested.

Back to the key questions

1. **When do people earn?** Labor income is concentrated roughly between 25 and 60, and is close to zero before 20 and after 65.
2. **When do they consume?** At *every* age, and far more evenly than they earn — which is why the gaps exist at all.
3. **How are the gaps bridged?** Only two channels: **transfers** and **asset-based reallocations**. Children are financed by transfers almost everywhere; for the old the mix is institutional — more public transfers in Europe, more assets in the United States.
4. **What does age structure imply for the macroeconomy?** Profiles $\times$ population gives the **support ratio**, and it falls as populations age. Whether that means *slower growth* is not settled: **no** negative relation across countries (Acemoglu & Restrepo), **yes** across US states (Maestas et al.).